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Steve Vercelli
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10 CFR 50.73

RBG-48053

October 29, 2020

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2020-004-00, "Inadvertent Initiation and Injection of High Pressure Core Spray (HPCS)"

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report. This document contains no commitments. If you have any questions, please contact Mr. Tim Schenk, Regulatory Assurance Manager, at 225-381-4177.

Respectfully,

A handwritten signature in black ink, appearing to read "Steve Vercelli", is written over a faint, larger version of the same signature.

SPV/djp

Enclosure: Licensee Event Report 50-458 / 2020-004-00, "Inadvertent Initiation Injection of High Pressure Core Spray (HPCS)"

cc: NRC Regional Administrator - Region IV
NRC Project Manager - River Bend Station
NRC Senior Resident Inspector - River Bend Station
Louisiana Department of Environmental Quality
Public Utility Commission of Texas

Enclosure

RBG-48053

Licensee Event Report 50-458 / 2020-004-00, "Inadvertent Initiation Injection of High Pressure Core Spray (HPCS)"



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollcts.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104). Attn: Desk alt: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

River Bend Station

2. Docket Number

05000

458

3. Page

1 OF 3

4. Title

INADVERTENT INITIATION AND INJECTION OF HIGH PRESSURE CORE SPRAY

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
09	03	2020	2020	- 004 -	00	10	29	2020	N/A	05000 N/A
									Facility Name	Docket Number
									N/A	05000 N/A

9. Operating Mode

1

10. Power Level

92

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☐ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER
Licensee Contact

Tim Schenk, Regulatory Assurance Manager

Phone Number (Include area code)

225-381-4177

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
E	NA	NA	NA	Yes	NA	NA	NA	NA	NA

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year
NA	NA	NA

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On September 03, 2020 at 2048 CT, River Bend Station (RBS) experienced a High Pressure Core Spray (HPCS) system inadvertent initiation and injection into the core while the station was at 92 percent power. Control Room Operators took manual control of Feedwater Level Control to maintain a stable reactor water level. Once it was verified that feedwater was not needed for level control, Operations manually closed the injection isolation valve and performed an override of the HPCS system. Closure of the HPCS Injection Valve rendered this system incapable of responding to an automatic actuation and was declared inoperable.

This report is made in accordance with 10 CFR 50.73(a)(2)(v)(D) for any event or condition that could have prevented the fulfillment of the safety function of structure or system that is needed to mitigate the consequences of an accident.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
River Bend Station		05000-	458	YEAR 2020	SEQUENTIAL NUMBER 004	REV NO. 00

NARRATIVE**EVENT DESCRIPTION**

On September 03, 2020 at 2048 CT, the High Pressure Core Spray System (HPCS) inadvertently initiated and injected into the reactor vessel causing a rise in reactor water level. No related activity was in progress at the time of the event. The Main Control Room (MCR) operators took the following immediate actions to restore normal reactor vessel water level:

- Ensured HPCS was not needed for reactor water level control
- Took manual control of HPCS and terminated injection
- Took manual control of the Feed-Water Level Control system (FWLC)
- Restored reactor water level to within the normal operating band
- Placed Feedwater Level Control (FWLC) back in automatic

The Division 3 Diesel Generator started automatically as expected with HPCS initiation and was subsequently secured by Operations. Upon initial investigation, the Control Room Operators observed that an entire row of HPCS trip unit circuit cards had failed low on the trip unit rack. When the back panel back door was opened, a strong acrid odor was present.

SAFETY ASSESSMENT

The actual consequence as stated was a High Pressure Core Spray (HPCS) inadvertent initiation and injection into the reactor pressure vessel. There were no other actual consequences to general safety of the public, nuclear safety, industrial safety or radiological safety for this event. The core was not adversely impacted by this event, and the option to initiate a manual SCRAM was always available to Operations. The HPCS pump was manually overridden, rendering the system INOPERABLE, but still available to mitigate the consequences of an accident if needed.

The MCR was able to take manual control of the RPV water level and return it to normal conditions. From a nuclear safety perspective, this event had no adverse impacts on the core. All core responses were as expected and remain bounded by the current licensing basis. Therefore, it has been determined there was no Safety System Functional Failure.

EVENT CAUSE

A trip unit card located on rack Z2 developed an electrical short and caused an amperage spike. This amperage spike caused voltage to lower across rack Z1. Once this voltage lowered beyond a certain point, an invalid Level 2 signal was generated and caused a HPCS actuation.

The voltage across rack Z1 fell below 22 VDC but remained above 19 VDC. When this happened, unit input voltage at the rack did not go low enough for the card rack to completely lose power, but it was low enough for the card to incorrectly interpret the low voltage as a low voltage signal from the field. Since the trip unit is programmed in reverse (meaning it acts on a low/de-energized voltage signal from a field transmitter), it still had enough power to be operational and was able to generate a false (or invalid) Level 2 signal.

CORRECTIVE ACTIONS

- Completed Action - As an immediate corrective action, replaced three failed trip unit cards along with the power supply.

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		YEAR	SEQUENTIAL NUMBER	REV NO.
River Bend Station	05000-458	2020	004	00

NARRATIVE**PREVIOUS SIMILAR EVENTS**

LER 2018-002

On April 26, 2018 at 15:31 CT, I&C technicians were restoring Reactor Vessel Water Level Transmitter BZ1-LTNO81C to service with the reactor operating at 100% power. During this transmitter restoration, the station experienced an invalid initiation of High Pressure Core Spray (HPCS) with injection into the reactor vessel.